Worksheet

On Determination of Point Load Strength Index of Rock

Worksheet No.:
Page.:

Information Supplied by Customer

Customer :
Contract No. :
Project :

Laboratory Information

Report No. :
Test Method : ASTM D5731-16
Test Procedure No. : STP06-RKT-D5731
Date of Sample Received :
Date Test Commenced :
Date Test Completed :
Test Location : Laboratory / On-site *
Ambient Temp. (°C)

Sample Details

Customer Sample No.				
Borehole no.				
Depth / Level * m / mPD *				
Sample Origin				
Sample Description				
Lab Reference No.				
Distance (L) of specimen				
Orientation of discontinuities				

Equipment ID	Stopwatch :	Upper platen :	Loadcell :
	Thermometer :	Lower platen :	Dial Gauge :

Test Result

Moisture Condition	*(Note 2)*					
Test type + direction	*(Note 3 & 4)*	+	+	+	+	
Distance between platens	**D** mm					
	Average mm					
Distance for penetration, **D'**	*(Note 7)* mm					
Width	**W** mm					
	Average mm					
Cross-section area, **A**	*(Note 8)* mm^2					
Equivalent core diameter, D_e	*(Note 9)* mm					
Failure Load, **P**	kN					
Point Load Index, I_s	*(Note 9)* Mpa					
Point Load Index, I_{s50}	*(Note 9)* Mpa					
Failure	Mode *(Note 6)*	a / b / d *	a / b / d *	a / b / d *	a / b / d *	
	Location					
	Type *(Note 5)*	j / m / NF *	j / m / NF *	j / m / NF *	j / m / NF *	

Notes:

(1) * Delete as appropriate

(2) Moisture condition: AR: As received; O: Oven dried; S: Saturated; L: Laboratory air dried at (°C)

(3) Test tpye: D: Diametral test on core; A: Axial test on core

(4) Test direction: //=Parallel to place weakness; ⊥=Perpendicular to plane of weakness; R=Random or unknown orientation

(5) Failure type: j: joint failure; m: Material failure; NF: Not failure

(6) Failure mode: Refers to Figure 4 of test procedure STP06-RKT-D5731

a: Valid diametral test; b: Valid axial test; d: Invalid core test

(7) If significant platen penetration occurs, record the value D' measures at the instant of failure.

(8) For axial test only.
When D' is not recorded: A = W x D
When D' is recorded: A = W x D'

(9) Calculation:
When D' is not recorded: Diametral test: $D_e^2 = D^2$
When D' is recorded: Diametral test: $D_e^2 = D \times D'$
$I_s = P / D_e^2$ Correction Factor $F = (D_e / 50)^{0.45}$ $I_{s50} = F \times I_s$

Tested by: Checked by: